12.3.4 Memory representation of 2-D array

Normally, the two-dimensional array can be viewed as a matrix. The conceptual view of a 2-D array is shown below:

int A[4][3];

A[0][0]	A[0][1]	A[0][2]
A[1][0]	A[1][1]	A[1][2]
A[2][0]	A[2][1]	A[2][2]
A[3][0]	A[3][1]	A[3][2]

In the above example, the 2-D array name A has 4 rows and 3 columns.

Like one-dimensional, the 2-D array elements are stored in continuous memory.

There are two types of 2-D array memory representations. They are:

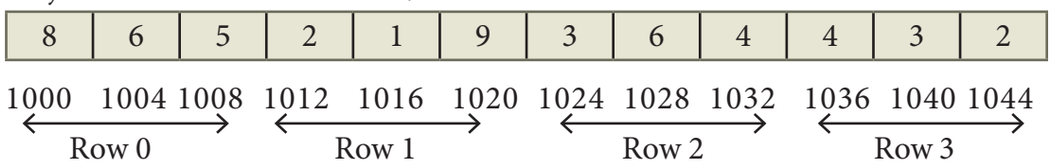
- Row-Major order
- Column-Major order

For example

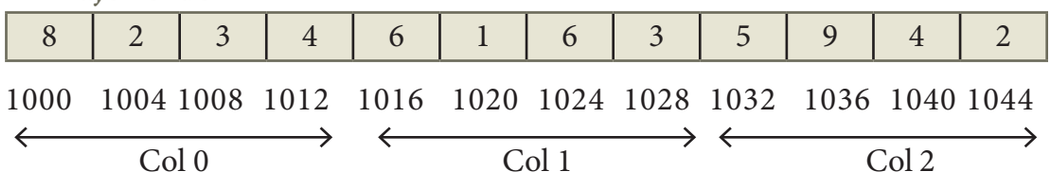
int A[4][3]={ { 8,6,5}, { 2,1,9}, {3,6,4}, {4,3,2} }

Row Major order

In row-major order, all the elements are stored row by row in continuous memory locations, that is, all the elements in first row, then in the second row and so on. The memory representation of row major order is as shown below;



Column Major order



12.4 Array of strings

An array of strings is a two-dimensional character array. The size of the first index (rows) denotes the number of strings and the size of the second index (columns) denotes the maximum length of each string. Usually, array of strings are declared in such a way to accommodate the null character at the end of each string. For example, the 2-D array has the declaration:

char Name[6][10];

In the above declaration, the 2-D array has two indices which refer to the row size and column size, that is 6 refers to the number of rows and 10 refers to the number of columns.

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